

Postulates: the basic rules we accept as true (no proving necessary)

Theorems: new rules we can prove are true by using the postulates (proving required)

To manually simplify boolean expressions, we collect both in a “toolbox” and use them to help us.

Referring back to last class, we introduced the Huntington Postulates:

Postulate	(a) OR / +	(b) AND / ·
1. Closure	closed under +	closed under ·
2. Identity	$x + 0 = x$	$x \cdot 1 = x$
3. Commutative	$x + y = y + x$	$xy = yx$
4. Distributive	$x + yz = (x + y)(x + z)$	$x(y + z) = xy + xz$
5. Complement	$x + x' = 1$	$x \cdot x' = 0$

There is an (a) column that gives the rules for the OR operator, and a (b) column that gives the rules for the AND operator. Each column can be derived from the other using the property of **Duality**.

Duality: take any true Boolean equation, swap every + with · and swap every 0 with 1, and the new equation is also true. (Leave the variables and their primes alone.)

Ex) Duality of $x + 0 = x$ becomes $x \cdot 1 = x$

Duality of $x + x' = 1$ becomes $x \cdot x' = 0$

These expressions are both true.

Below are the **Theorems of Boolean Algebra**. They are proven using the postulates:

Theorem	Statement	What it means
1	$x + x = x, \quad x \cdot x = x$	a value combined with itself is just itself
2	$x + 1 = 1, \quad x \cdot 0 = 0$	1 wins an OR; 0 wins an AND
3 (involution)	$(x')' = x$	two NOTs cancel out
4 (associative)	$x + (y + z) = (x + y) + z$ $x(yz) = (xy)z$	grouping doesn't matter
5 (DeMorgan)	$(x + y)' = x'y'$ $(xy)' = x' + y'$	flip each variable <i>and</i> swap the operation
6 (absorption)	$x + xy = x, \quad x(x + y) = x$	x “absorbs” the extra term

We can prove a theorem in two different ways: using a truth table or proving it with the postulates. Let's say we wanted to prove Theorem 1: $x + x = x$ using the postulates:

$$\begin{aligned}
 x + x &= (x + x) \cdot 1 && \text{Postulate 2: anything } \cdot 1 \text{ is itself} \\
 &= (x + x)(x + x') && \text{Postulate 5: replace the 1 with } x + x' \\
 &= x + xx' && \text{Postulate 4: factor (distributive)} \\
 &= x + 0 && \text{Postulate 5: } xx' = 0 \\
 &= x. && \text{Postulate 2: anything } + 0 \text{ is itself}
 \end{aligned}$$

DeMorgan's Theorem: tells us how to take the complement of an expression using two rules:

1. Flip each variable (prime to not prime, not prime to prime)
2. Flip each operator (OR to AND, AND to OR)

Ex) Find the complement of $F = x + y'z$

$$\begin{aligned} F' &= (x + y'z)' = x' \cdot (y'z)' && \text{(swap the outer OR to AND; flip } x\text{)} \\ &= x' \cdot (y + z') && \text{(swap the inner AND to OR; flip } y' \text{ and } z\text{)} \\ &= x'(y + z'). \end{aligned}$$

Ex) Find the complement of $F = AB' + C$

$$\begin{aligned} F' &= (AB' + C)' = (AB')' \cdot C' && \text{outer OR becomes AND} \\ &= (A' + B) C' && \text{since } (AB')' = A' + (B')' = A' + B \end{aligned}$$

With no parenthesis to guide you, evaluate in this order:

(1) parentheses $\rightarrow$ (2) NOT $\rightarrow$ (3) AND $\rightarrow$ (4) OR

Ex) Simplify $F = A'BC + A'BC' + AB$

$$\begin{aligned} F &= A'B(C + C') + AB && \text{factor } A'B \text{ out of the first two terms} \\ &= A'B \cdot 1 + AB && \text{Postulate 5: } C + C' = 1 \\ &= A'B + AB && \text{Postulate 2: } A'B \cdot 1 = A'B \\ &= B(A' + A) && \text{factor } B \\ &= B \cdot 1 = B. && \text{Postulate 5 then Postulate 2} \end{aligned}$$

Ex) Without using a truth table, confirm that $ab' + a'c + b'c = ab' + a'c$

$$\begin{aligned} b'c &= b'c \cdot 1 \\ &= b'c(a + a') \\ &= ab'c + a'b'c \\ ab' + a'c + b'c &= ab' + a'c + ab'c + a'b'c \\ &= (ab' + ab'c) + (a'c + a'b'c) \\ &= ab'(1 + c) + a'c(1 + b') \\ &= ab' \cdot 1 + a'c \cdot 1 \\ &= ab' + a'c \end{aligned}$$